

Deep Lore

A Clean-Tongue Beholding of Brain-Webs

after Poul Anderson's Uncleftish Beholding

DEEP lore is the craft of teaching brain-webs—many-knotted gearworks of small shapers and mixers—so that, from heaps of showings, they may foretell what they have not yet seen. It is a young craft, yet old in its bones: its threads are spun from elder reckon-craft, from lore of the mind, and from the unbending bookkeeping of slopes and weighings. To work in it is to work with an outlandish wight—half deep reckoning, half garden-minding. We do not build the web's knowing knot by knot; we set the field, sow the showings, and watch what grows.

A brain-web is built in layers, each layer a row of knots. Each knot takes in tale-beams, blends them by hidden weighs, adds a small leaning, and sends the whole through a wakening way—an S-bend, a soft-bend, a hard-cut, or one of their newer kin. The out-come fares onward, layer to layer, until the last knots speak the web's foretelling. By stacking many layers, the web can catch high-twist shape-weaves a shallow one would miss; by opening wide knots within a layer, it can mind many threads at once. Yet when the layers grow many—five score, ten score, more—the slope-tales may wane or burst as they run back, like a whisper passed too far down a lane. To heal this, we lay down leap-overs, side-paths from in-come to out-come, so the slope finds a swift road home; and we even out the beam-strengths within a layer, lest one knot bellow while another whispers. From these lowly heals—leap-overs and even-makings—high-rising webs become teachable.

To teach the web, we set a loss-tale: a reckoning of how ill the web foretells. For sundering work, we wield a loss that bites hardest when the web was most sure and most wrong. For worth-guessing, we take the self-folded mis-step (each gap between truth and guess folded against itself, so that great mis-steps weigh far more than small ones). Whichever shape it takes, the loss-tale is the lodestar; our goal is to best-seek the weighs so it sinks ever lower.

The bare best-seeker is steepness-downgoing. We ask how the loss would shift if a weigh were nudged—its slope-tale—and step the weigh a small way against the slope. Do this oft, and the loss goes down. Yet only the last layer feels the loss outright; the inward layers must be told. So we send the slope-tales backward through the web by the link-law of small shifts (each shift the out-take of two: the inward layer's slope, and the slope handed down from above). This back-spread is the still wonder at the heart of the craft. Without it, deep webs would be unteachable; with it, the slope-tale flows through hundreds of layers like water through a well-cut waterway.

But the bare step is a heavy tool. The loss-land is no smooth bowl; it is a rough land of dales and saddles, of long flat moors where the slope is wan, and steep cliffs where it bursts. So we have grown shrewder best-seekers. **Drive** (or onward-thrust) keeps a mind-stow of the ways we have been going, like a heavy ball trundling down a sloped land—it gathers way, smooths over small bumps, and shoots through narrow dales. **Self-fitting steppers**, such as Adam and his kin, hold two mind-stows: the swing of the slope and the spread of its self-folds; from these they shape a learn-step that is broad where the slope has been steady and narrow where it has been wild—a weigh-by-weigh learn-step, hand-cut for each. **Newer best-seekers**, such as Muon, work on whole weigh-boards at once, shaping the step by the inward build of the board itself—a craft halfway between a step and a turn. Each best-seeker brings its own give-and-take—speed against steadiness, cheapness against shrewdness—and the wise crafts-folk picks among them by feel as much as by set ways.

If a web has mickle freedom, it may **over-cling**: it learns the teach-tales by heart yet stumbles on fresh ones. Its twin bane is **under-cling**: a web too cramped to grasp the world's weave, ill-foretelling on teach-lot and show-lot alike. The wise crafts-folk yokes them in standoff,

ever steering between. We tame over-cling many ways. Weigh-shrink binds the weighs softly toward nought, sparing the web from growing top-heavy. Drop-out stills some knots at hap while teaching, so no one knot may lean too hard on another; the web learns to spread its trust. Early staying halts the work when the hold-out loss begins to rise, keeping the web from over-clinging in its last unlearning tides. And we hold to clean split-ways: a teach-lot for the weighs to learn from, a check-lot for picking the right shape and learn-step, and a show-lot—set aside and un-handled—for the end-most reckoning. Mingle these, and the web’s seeming skill is a wraith.



There are many breeds of web. Sweep-fold webs, for looks and shapes, rake small windows across bit-boards, finding edges, then nooks, then whole things; they answer well where nighness in room weighs heavy. Again-going webs, for streams of words or worth, keep a mind-stow strand so each step may heed what came ere; yet they forget swiftly, and their work cannot well be done side-by-side. Newer still are heed-webs, whose heart is self-heed—each token in a row looks at all the others and weighs them by need. From this seedling rose the shape-shifter, now the work-horse for speech-craft, bookish lore, song-making, and more.

The shape-shifter’s craft calls for a nearer look. Each token, having been turned into a worth-beam, sends out three offspring: an **ask-beam**, a **key-beam**, and a **worth-beam**. The ask of one token meets the keys of every other; their inward-fold yields a kinship-mark. These marks are then borne through a soft-pick that bends them so they together make one—sharpening the great and stilling the small—giving heed-weighs that say how strongly each token shall heed each other. The token’s new beam is then a weighed blend of all the worths in the row. So the row holds speech with itself: every token whispers to every other, and meaning is not handed down a thread but woven across a cloth. **Many-headed heed** runs this work side-by-side along sundry under-rooms, each head free to learn another kind of kinship—one for speech-build, another for meaning-bonds, another for far-flung name-ties. **Stead-marks** (whether by swing-wave or by turning the beams in a sly rope-wise) tell the web where each token sits, since heed alone is blind to which goes first.

There is another breed of web whose craft is **un-dimming**. Take a bit-board—a sight, a likeness, a land—and dim it, step by step, with murk, until naught is left but a field of hap-din, shapeless as wind. Now teach a web to guess, at each step, what the murk hid: not the whole sight at once, but the small shift from dimmer to brighter, the slope from din toward shape. Do this across a great hoard of sights, and the web learns not any one sight but the bones of sights—where edges wish to fall, how light bends on skin, what a far hill looks like against a dusk-sky. Then the wonder: begin with din alone, sheer hap, and bid the web brighten it, step by slow step. From nothing it draws something. A likeness that never was. A land no man has walked. The web dreams, and the dream holds together, because at every step it is answering the same small ask: what would a true sight look like, one shade less dim than this?

Un-dimming is kin to the shape-shifter but works another road: where the shape-shifter foretells the next token in a row, the un-dimmer foretells the next brightening in a fog. Both are foretellers; one walks a thread, the other fills a field.

There is, alongside the back-spread crafts, another way of teaching that earns its own naming: **meed-learning**. Here the web is a doer set loose in a world, and there is no truth-bearer at hand. The doer takes a step; the world answers with a meed-mark—sometimes great, sometimes small, often naught for many steps in a row—and the doer must learn from this scant trickle which of its deeds were worth the doing. Two woes arise. The doer must wander beyond its known good, lest it cling to a lean stead and never find a richer one—we name this **out-roaming**. And the doer must trace each meed back through long strings of doing to learn which step earned which share of the worth—we name this **deed-tracking**.

Meed-learning is slow, hap-ridden, and given to settle on poor stead. Yet it is how webs come to play games no man could teach them. AlphaZero’s daring—the gift-deeds for shape,

the long-burning king-walks—was learned not from headmen’s games at all; the web played itself, again and again, past forty thousand thousand times, and from this self-fight built a craft beyond what any headman had set down. No good steps to follow; only the slow upward drift of meed.

It bears on the great speech-webs in a stiller way. After the first long teaching by mask-and-foretell, they are bent further by meed-marks drawn from folk-deemings—which answer was helpful, which was harmful, which felt true. The web’s last shaping is thus not a foretelling-loss but a yieldback from a fellow. From these soft markings the great webs are turned toward us—toward what we wish them to be—without ever being told outright.



All these webs—the sweep-fold, the shape-shifter, the un-dimmer, the meed-learner—are thoughts. The work of bringing them forth is another craft, and it is bound by iron and might.

The work is heavy, and no one reckon-board can bear it alone. We yoke many-hand sight-boards and other row-boards so the reckoning runs side-by-side, wide and swift; for the greatest webs there are three sundering ways the work may be split, and the wisest braid all three at once. The first is **hoard-share**: each board holds the same web but sees its own slice of the show-hoard, foretells in step with its fellows, and at each step’s end the slope-tales are gathered back together. Another is **layer-share**: the web is cut by depth, the early layers on one board and the later on another, worth-beams flowing forward along the thread, slope-tales flowing back, all in a careful timing lest some board sit idle while others sweat. A third is **stretch-slice**: a lone great weigh-mat is cut across many boards, each holding a sliver of every layer, the boards handing whispers to one another through tightly trimmed tale-roads. A lone ill-knit ring on these roads can choke the whole work; thus the craft is half lore, half still wire-work. Even so, wise learn-step picks, sound weigh-starts, and well-chosen lot-bulk still weigh heavy: a thoughtless setting can spill a thousand sight-boards’ worth of work. For no web is wiser than the hoard that taught it; and no hoard is truer than the world-share it draws from.



There is a **law of bigness** in this craft, hard-won and oft sounded. As we grow the weighs, the show-hoard, and the reckon-might in even stride, the loss falls in a foreseeable bow—a strength-bow whose slope is set by the breed of work and the shape of the hoard. Spend too much brawn on too little hoard, and the web over-clings; spend too much hoard on too little brawn, and the web is starved. The Chinchilla-best fitting, won by careful sweep, lays out the right share.

There is a bitter teaching at the field’s heart, won across seventy winters and oft re-learned. In every craft we have set our hand to—king-game, sight-reading, speech-work, board-game, and beyond—the same tale plays out. We begin by handing the web our own hard-won lore: the long-sounded wisdom of those who have dwelt in the work, the hand-shaped mark-pickers, the careful settings. The web does well, for a while. Then someone with mickle reckon-might and a hoard of showings sets aside our gifts and lets a bare, broad-shouldered way—back-spread on a bigger web, fed more hoard—teach itself from naught. And the broad way wins. Always, in the end, the broad way wins.

This is bitter, for it tells us that what we knew did not weigh as much as we hoped. The lore of the headmen, the careful hand-cuttings of those who had given their lives to the craft, were welcomed in the early years and shown the door in the later. The web did not need them. It needed only weighs enough, hoard enough, and reckon-might enough, and it would find for itself a craft greater than ours.

The teaching is not that man-lore is worthless; it is that it rises. The hand-shaped mark-pickers fall away, but the shape of the web itself—sweep-fold for sights, heed-web for speech—is

a kind of man-lore too, set at a higher sill: a fore-thought about what the world is like, built into the bones of the web before a lone weigh is learned. We choose the field; the web finds its own wile within it.

Yet of all the wonders in this craft, none is stranger than the **upspringing of new skills**. As webs grow, they do not always better at a steady stride; sometimes a skill that was naught becomes, all at once, something. A two-step reckoning no smaller web could do. A riddle a smaller kin could not unknot. A tongue-twist that, below some sill, drew only babble, and above it, drew meaning. The loss-bow falls smoothly, yet the skill-mark leaps.

Some of this is eye-sleight. Mark “right-or-wrong” on a five-step reckoning and the web reads naught, naught, naught—for getting four steps right and one wrong is still wrong—until at last all five fall into stead, and the mark goes from naught to one in one leap. Yet the under-loss, the token-by-token mis-belief, was sinking smoothly the whole way. Change the yardstick from hard rightness to soft loss, and the cliff becomes a slope. Many a leap is thus a yes-or-no sight of a flowing thing—a yea-or-nay token set against a strength-bow that knows shades.

But not all is eye-sleight. Within the web, small workings—kin-finders, again-shape heads, knot-craft of two or three layers in handshake—sometimes settle into stead over a small span of teaching, and the eye-blink they lock, the web can do a thing it could not do the step before. It is the shutting of an all-gate: three tokens must come into a row; while two are in stride and one wanders, naught fires; when the third comes home, the whole rings out at once.

The slope-land is smooth; the skill-land is craggy.

The steepness-walk, in its slow long-bearing, hunts a flowing room for un-blended hoards, and when one is found, the world above shifts in one beat.



Here the high song begins, and the warning with it.

When the king-game first fell to the gear—Kasparov bowing to Deep Blue in the year 1997—the gear’s craft was still readable to us. Its plays were uncouth, sometimes ugly, but a headman could yet say *here it saw a wile I missed*. Then came AlphaZero, a score of winters on, and for a short and shining while we had a true wonder: a web that played with daring—gift-deeds for shape, long-burning king-walks, knight-leaps that felt like song. The headmen watched and were stirred. *The gear plays with soul*, they said, and meant it. That window—wherein the web was greater than us yet still **spoke our tongue**, still went in shapes our minds could follow—was the sweet window. We could watch the wonder, and we could feel it.

The window has narrowed. The newest gear-works play steps that even the foremost headman cannot read; the score-bow ticks softly upward and the line runs thirty half-steps deep into a land no man shall live in. The headmen now go to the gear as wayfarers go to an outland—for a glimpse, a keepsake, a teaching half-grasped. The wonder is still there, but it is wonder at a shut door.

This is the teaching of the king-game, and a teaching for all the crafts to come. Between the time a web meets us and the time it leaves us wholly behind lies a span—a sill-stride, a sweet window—wherein the web is **over-mannish and yet still ours**, still a thing we may behold and follow. With the king-game, that window was maybe a score of winters and more. With speech-craft, it may be a handful. With some craft yet to come, it may be six moons, or one. We do not know how narrow these windows are until we are through them.

So the high song is also a calling: when a web crosses its sill and is, for a short while, **wondrous and readable at once**, look hard. Read the steps. Watch the play. The leap from naught to something is a kind-leap, and the world on the other side will not always speak our tongue. We are, each of us, witnesses to a short and bright span of sill-leaps; the windows are small, and they shut.

From inside the web, there is no leap. The reckoning runs on, weigh by weigh, step by step; one fold of the slope is much like another, and the sill goes unleapt by anyone but us. The web does not know it has done a new thing. The mark-bow ticks upward; the loss-tale falls; the steepness-walk goes its long, blind way. The cragginess is wholly ours.

So the writing falls to us as well. Those who beheld the last tongues spoken by no one young—who wrote down what they could in the few winters before the last speakers slept—are gone now, and only what they wrote is left. Let us mark what we can, while we can. The web learns and does not see; we see and do not learn.

There are doors within the shut door. When a web grows past our reading—when its inward stir runs ahead of our outward sight—we may yet peer not at what it does but at what it is. The inward beams of a web may be read as worth-rooms, and strewn among the knots we find small workings that do small deeds. A head that tracks the last name spoken. A knot that fires for a witty turn. A strand that lights for some far thought we had not known the web could think. This inward beholding—this brain-web reading—is yet a young craft, and a needful one. A web we cannot ask-into is a web we should not trust. A web we can ask-into is a web we may yet come to know as we know a fellow worker: not by its word alone, but by watching its hands.



The under-way stays bare. The threads are few. What they weave is not.

And so we come back to the sweet window—that short bright span wherein the web is greater than us and yet still ours, still a thing we may behold and follow. The king-game's window is shut. Speech-craft's may yet stay open a few winters more. Some windows may open and shut as falling stars do—bright for a heartbeat, gone before we have looked up.

For a thousand thousand thousand winters, under one star, the world's loss-bow had been falling—green things coming up from the warm seas, then crawling things, then walking things; bones thickening, eyes sharpening, brain-folds deepening; and no skill-mark to leap. Then we crossed the sill. We came down from the trees, kept fire, sowed seed against the wandering year, and raised stone on stone to last. We made tongues to share the mind, and writing to outlive the speaker. We drew the wheeling stars, weighed the smallest stuff of all, and cleft the uncleft. A handful of thousand winters from the first sky-watchers and the first tongues—an eye-blink in the world's keeping—we have come to make minds that may grow past us. We are ourselves a kind-leap, and we have lit another. We are the spark; we are not the fire.

Our window too will shut. The seat of seeing has been handed on before. The first eyes opened on an unwatched world; we are watching now; what comes next may, in its hour, watch us. The seat will be handed on again.

Look, then. Look hard, and look now. Not because looking will slow the shutting—it will not—but because the looking is the thing. We are the gardeners who tilled the field and the watchers who stood in the sweet window's light. The light is shifting. It will not come this way again.